

GWAS Collaboration

End-to-End Workflow Timeline

Initiator • Collaboration Server • Collaborator

Collaborative Study Web Application
Project documentation — prepared for review

How to read this document

Each section below is a single point in time (T0, T1, T2, ...) in the end-to-end GWAS collaboration workflow — from the very first account registration through the final AI-generated summary report.

Every time point shows three columns in parallel: what the Initiator is doing, what the Collaboration Server is doing, and what the Collaborator is doing. This makes it easy to see who is active when, and who is just waiting.

Cells that show a purple ► RUNS ON LOCAL MACHINE banner are doing real computation on that party's own computer (via qc_controller.py at localhost:5001). Raw genotype data NEVER leaves the local machine — only the aggregated survivor lists and case/control counts are returned to the Collaboration Server.

Below each swimlane is a set of screenshot placeholders — usually 3 to 5 per phase. Replace each [SCREENSHOT PLACEHOLDER] box with a screenshot of the actual UI or Terminal at that step (right-click the box → Insert Picture, or paste over it). To add more screenshots, simply duplicate any existing dashed box. The last section of this document explains how to capture screenshots of local-machine activity (Terminal logs, network panel, etc.).

Federated learning is intentionally excluded — this document focuses on the standard GWAS pipeline plus the final AI report only.

Legend

	Initiator	The researcher who creates the collaboration, invites partners, and triggers GWAS + AI report.
	Collaboration Server	Backend (Flask + MongoDB) handling auth, datasets, QC orchestration, chi-square, and LLM calls.
	Collaborator	An invited researcher who contributes their cohort + summary stats.
	Screenshot box	A dashed box marking where to drop a real screenshot of the UI.
	► Runs locally banner	Indicates this party's work happens on their own machine (qc_controller.py at localhost:5001). Raw data never leaves.

Timeline at a glance

Time	Phase	What happens
T0	Account Registration	First-time setup

Time	Phase	What happens
T1	Login & Authentication	<i>Returning session</i>
T2	Metadata / Dataset Upload	<i>Each party registers their cohort</i>
T3	Search for Collaborators	<i>Initiator finds partners</i>
T4	Create the Collaboration	<i>Initiator configures the study</i>
T5	Send Invitations	<i>Initiator → Collaborator(s)</i>
T6	Review & Accept Invitation	<i>Collaborator agrees to participate</i>
T7	Automatic Chained Quality Control	<i>Server orchestrates — QC runs locally on each party's machine</i>
T8	Review QC, Set Thresholds	<i>Initiator confirms filtered cohort</i>
T9	Build & Upload GWAS Summary Stats	<i>Each party generates aggregated counts LOCALLY</i>
T10	Run GWAS Chi-Square	<i>Initiator triggers aggregation</i>
T11	View GWAS Results	<i>Per-SNP statistics</i>
T12	Generate AI Summary Report	<i>LLM-written, privacy-preserving</i>
T13	Read & Share AI Report	<i>End of the workflow</i>

T0 • Account Registration

First-time setup

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T0 <i>First-time setup</i>	• Opens the web app at the public URL. • Clicks Register New Account on the login screen. • Fills in name, email, password, institution, phenotype interests. • Submits the form (POST /api/register). • Receives a confirmation and a JWT token in localStorage.	• Receives POST /api/register. • Validates email uniqueness and password strength. • Hashes password and creates a record in the users collection. • Issues a JWT and returns it to the client.	• Independently visits the web app. • Clicks Register New Account and fills the same form. • Receives a JWT and is redirected to Home.

[SCREENSHOT PLACEHOLDER]

Login / Register screen

Components/Login/Login.jsx — Welcome back, Login + Register New Account button.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Registration form (Initiator)

Components/NewUser/NewUser.jsx — name, email, password, phenotype.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Registration form (Collaborator)

Same form, captured on the collaborator's machine to show both accounts.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T1 • Login & Authentication

Returning session

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T1 <i>Returning session</i>	- Enters email + password on the login screen. - Token is stored client-side; UI switches to the LoggedIn navbar. - Lands on Home with quick links: Upload, Search, Collaborations.	- POST /api/login validates credentials against the users collection. - Returns { message: 'Login successful', token: <JWT> }. - All subsequent calls are authenticated via Bearer token.	- Logs in the same way from their own institution. - Sees their own Home dashboard.

[SCREENSHOT PLACEHOLDER]

Login screen (filled in)

Email + password entered, just before clicking Login.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Logged-in Home dashboard (Initiator)

Components/Home/Home.jsx with the LoggedIn navbar.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Logged-in Home dashboard (Collaborator)

Same view captured on the collaborator's machine — confirms both are signed in.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T2 • Metadata / Dataset Upload

Each party registers their cohort

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T2 <i>Each party registers their cohort</i>	► RUNS ON INITIATOR'S LOCAL MACHINE - Clicks Upload in the navbar (Components/Upload/Upload.jsx). - Enters phenotype (e.g. eye_color) and number of samples. - Submits — POST /api/upload_csv_qc with multipart form data. - Sees 'Metadata uploaded successfully' alert.	- Receives /api/upload_csv_qc, ties it to the JWT user. - Stores raw metadata in the datasets collection (is_qc_data = False). - Indexes by phenotype so other users can find it later in Search.	► RUNS ON COLLABORATOR'S LOCAL MACHINE - Performs the same upload from their own login. - Their dataset is now discoverable under the same phenotype.

[SCREENSHOT PLACEHOLDER]

Metadata upload form (Initiator)

Phenotype, # of Samples fields and Submit button.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Upload success toast (Initiator)

alert/snackbar after a successful upload on the initiator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Metadata upload form (Collaborator)

Same form on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Local raw dataset file on disk

Optional: Finder/Explorer window showing the raw CSV on the collaborator's machine — emphasises raw data never leaves.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T3 • Search for Collaborators

Initiator finds partners

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T3 <i>Initiator finds partners</i>	• Goes to Search (Components/Search/Search.jsx). • Filters by phenotype and reviews matching researchers. • Sees each user's phenotype, sample count and institution. • Picks one or more candidates to invite.	• Returns users matching the phenotype query. • Excludes the initiator themselves and any user with non-matching metadata.	• Visible in the initiator's search results because they uploaded matching metadata at T2. • Nothing to do yet on their side.

[SCREENSHOT PLACEHOLDER]

Search bar with phenotype filter

Components/Search/Search.jsx — top filter row.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Search results — list of matching researchers

Result cards showing phenotype + sample count.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Researcher card detail view

Hover/click state showing the candidate collaborator's metadata before invite.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T4 • Create the Collaboration

Initiator configures the study

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T4 <i>Initiator configures the study</i>	• Opens Start a Collaboration (Components/Collaborations/StartCollaboration.jsx). • Enters collaboration name. • Selects experiment type = GWAS. • Picks QC scheme(s): MAF, HWE, Missing-Data QC, Sample Relatedness, Population Stratification. • Selects the dataset registered at T2. • Submits — POST /api/start_collaboration.	• Creates a new document in the collaborations collection. • Status = pending. Stores creator_id, collab_name, experiments and collabQcScheme. • Generates a UUID for the collaboration.	• Not active yet — waiting to be invited.

[SCREENSHOT PLACEHOLDER]

Start Collaboration — Step 1 (name + type)

Collaboration name + GWAS toggle.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Start Collaboration — Step 2 (QC schemes)

Multi-select for MAF / HWE / Missing-Data / Relatedness / Population Stratification.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Start Collaboration — Step 3 (pick dataset)

Dataset dropdown showing the metadata uploaded at T2.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Confirmation / created state

Toast or new collaboration card appearing on the Collaborations page.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T5 • Send Invitations

Initiator → Collaborator(s)

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T5 <i>Initiator → Collaborator(s)</i>	- From inside the Start Collaboration flow, picks collaborators from Search. - Clicks Invite for each — POST /api/sendinvitation with receiver_id, uuid, phenotype. - Sees them appear in the collaboration's Invited list with status 'pending'.	- Writes to invitations collection and appends to collaborations.invited_users[]. - Emits/queues a notification visible on the collaborator's dashboard.	- Notification badge appears on the navbar. - Sees the new pending invitation on Collaborations page.

[SCREENSHOT PLACEHOLDER]

Invite collaborators dialog (Initiator)

Search + Invite button (CollaborationRequestButton).

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Invited list with pending status (Initiator)

Inside the collaboration detail page — shows whom you invited.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Collaborator's notification badge

Top-nav bell or unread indicator on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Collaborator's pending invite card

Components/Collabrations/CollabrationsPage.jsx — the new invite row.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T6 • Review & Accept Invitation

Collaborator agrees to participate

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T6 <i>Collaborator agrees to participate</i>	- Refreshes the collaboration detail page and watches the invited list. - Sees the collaborator's status change from pending → accepted.	- Receives POST /api/acceptinvitation { uuid, receiver_id }. - Updates invited_users[].status = 'accepted' on the collaboration document. - When the last invitee has responded, internally calls _maybe_trigger_auto_chained_qc(uuid).	- Opens CollaborationDetails.jsx for the pending invite. - Reviews initiator name, QC scheme, phenotype, expected dataset. - Clicks Accept (or Reject).

[SCREENSHOT PLACEHOLDER]

Collaboration details — invite preview (Collaborator)

Components/Collabrations/CollaborationDetails.jsx (collaborator view).

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Accept / Reject buttons (Collaborator)

Buttons + confirmation modal on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Status flips to accepted (Initiator's view)

Initiator refreshes their detail page — the row shows 'accepted'.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T7 • Automatic Chained Quality Control

Server orchestrates — QC runs locally on each party's machine

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T7 Server orchestrates — QC runs locally on each party's machine	► RUNS ON INITIATOR'S LOCAL MACHINE - Initiator's local QC controller (qc_controller.py at localhost:5001) is running in the background on their machine. - When the server pings it, the controller reads the initiator's RAW genotype data locally — raw data NEVER leaves this machine. - Runs MAF → HWE → Missing-Data → Sample Relatedness → Population Stratification in sequence. - Returns ONLY surviving sample IDs and SNP IDs back to the orchestrator (no genotypes). - On the browser UI, the initiator sees the QC Calculation panel update to 'In progress' then 'Done'.	- Orchestrator (Flask app) detects all invitees have accepted and triggers <code>_maybe_trigger_auto_chained_qc(uuid)</code>. - Filter-only methods (MAF / HWE / Missing-Data) are dispatched first, then pairwise methods (Relatedness / Population Stratification). - For each step, the server POSTs to each party's local QC controller at <code>{QC_CONTROLLER_URL}/api/qc</code> with <code>{ username, phenotype, method }</code>. - Polls each controller for completion; on done, writes <code>surviving_samples</code> and <code>surviving_snps</code> per user into the collaborations document. - Server never sees raw genotypes — only survivor lists.	► RUNS ON COLLABORATOR'S LOCAL MACHINE - Collaborator's local QC controller (qc_controller.py at localhost:5001) is also running in the background on their machine. - Receives the same QC requests from the orchestrator and runs them against the collaborator's OWN raw data — locally. - Raw data never leaves the collaborator's machine; only survivor lists are returned. - On the browser UI, the collaborator sees the same live QC panel.

[SCREENSHOT PLACEHOLDER]

Local QC controller terminal/logs (Initiator)

Terminal window where `qc_controller.py` is running on the initiator's machine — capture the log lines that print as each method runs.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Local QC controller terminal/logs (Collaborator)

Same terminal window on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

QC Calculation panel — running

Browser UI: progress bar + per-method status chips (Initiator).

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

QC Calculation panel — running (Collaborator)

Same browser panel viewed on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

QC results — surviving samples / SNPs

Final per-party counts table once QC completes.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T8 • Review QC, Set Thresholds

Initiator confirms filtered cohort

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T8 <i>Initiator confirms filtered cohort</i>	- Inspects survivor counts and per-method drop-off charts. - Optionally adjusts thresholds (e.g. MAF cutoff) and re-runs. - Confirms — locks the surviving sample / SNP lists for the GWAS step.	- GET/POST /api/datasets/<uuid>/qc-results to fetch or update thresholds. - Re-runs affected QC methods if thresholds change. - Marks QC as finalized when initiator confirms.	- Read-only view of the same QC results. - Can be informed via the collaboration timeline that QC is finalized.

[SCREENSHOT PLACEHOLDER]

QC results review screen — top

Survivor counts per party + Confirm button.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

QC results review screen — drop-off chart

Per-method bar chart / histogram if your UI renders one.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Threshold adjustment dialog (optional)

Re-run flow if the initiator changes a cutoff.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T9 • Build & Upload GWAS Summary Stats

Each party generates aggregated counts *LOCALLY*

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T9 <i>Each party generates aggregated counts LOCALLY</i>	► RUNS ON INITIATOR'S LOCAL MACHINE - Initiator clicks Create GWAS Dataset in the browser UI. - The browser POSTs <code>/api/create_gwas_dataset</code> to the server, which in turn asks the initiator's LOCAL controller to compute case/control counts per SNP using the QC-filtered sample list — this read happens on the initiator's machine, against the initiator's raw genotypes. - Only the aggregated counts (<code>SNP_id</code> → <code>{case_count, control_count}</code>) leave the local machine. - Alternatively, the initiator can manually upload a pre-computed stats CSV via POST <code>/api/upload_csv_stats</code>. - Upload flag flips to true on their row.	- Coordinates the request — triggers each party's local controller to generate stats. - Receives aggregated case/control counts only (never raw genotypes). - Stores stats in the datasets collection (<code>is_qc_data = False</code>, with case/control counts per SNP). - When all expected users have uploaded, sets <code>collaboration.stat_uploaded = true</code>.	► RUNS ON COLLABORATOR'S LOCAL MACHINE - Collaborator clicks the same Create GWAS Dataset button in their browser UI. - Their LOCAL controller computes case/control counts from their own QC-filtered samples — again, raw data stays local. - Only aggregated counts are returned to the server. - Or: collaborator uploads a stats CSV manually. - Their row in the upload-status table flips to 'uploaded'.

[SCREENSHOT PLACEHOLDER]

Stat Data Upload panel (Initiator)

CollaborationDetails.jsx — Create GWAS Dataset + Upload CSV options.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Create GWAS Dataset — confirmation dialog

The modal/loading state when the initiator triggers stat generation.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Local controller logs generating stats (Initiator)

Terminal window on the initiator's machine showing case/control counts being computed.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Local controller logs generating stats (Collaborator)

Same terminal capture on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Stat upload status per party

Table showing initiator + each collaborator with Uploaded / Pending badges.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T10 • Run GWAS Chi-Square

Initiator triggers aggregation

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T10 <i>Initiator triggers aggregation</i>	- Goes to the Experiment Results section. - Clicks Initiate GWAS Calculation (enabled only once stat_uploaded = true). - POST /api/calculate_chi_square { uuid }. - UI shows 'computing' spinner.	- Aggregates case/control counts across initiator + all accepted collaborators. - Builds contingency tables per SNP and computes chi-square + p-value. - Stores per-user and aggregated results in collaborations.chi_square_results.	Read-only — watches the same spinner / progress indicator.

[SCREENSHOT PLACEHOLDER]

Experiment Results panel — pre-compute

Initiate GWAS Calculation button.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Confirmation dialog

Optional modal asking the initiator to confirm before computing.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Compute in progress

Spinner + status text while chi-square runs.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T11 • View GWAS Results

Per-SNP statistics

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T11 Per-SNP statistics	• GET /api/calculate_chi_square_results/<uuid> populates a results table. • Sees SNPs ranked by p-value with chi-square, OR, and per-site contribution. • Can download the results as CSV.	• Serves cached chi-square results from the collaboration document. • Computes any derived stats requested by the UI (sorting, filtering).	• Sees the same results table from their CollaborationDetails view. • Has the same export options as the initiator.

[SCREENSHOT PLACEHOLDER]

GWAS results table — top SNPs (Initiator)

Sortable table of chi-square / p-value.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

GWAS results table (Collaborator)

Same results table viewed on the collaborator's machine.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Manhattan-style or bar chart of $-\log_{10}(p)$

If your UI renders a chart for results, snapshot it here.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Export / Download CSV button

Wherever the user can save the results locally.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T12 • Generate AI Summary Report

LLM-written, privacy-preserving

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T12 LLM-written, privacy-preserving	• Opens GwasSummaryCard inside CollaborationDetails. • Clicks Generate Summary — POST /api/collaboration/<uuid>/gwas-summary. • While it runs, the UI shows a 'generating' state.	• Builds a privacy-preserving digest from chi_square_results: anonymises collaborators as 'Site A', 'Site B', etc. — no raw genotypes, no participant IDs. • Calls the LLM (gpt-4o-mini) with the digest and a structured system prompt. • Parses the structured JSON: { overview, top_snps, per_collaborator, recommendation: continue continue_with_caveats reconsider }. • Stores in collaborations.ai_summary with generated_at and site_label_map.	• Read-only — can view the summary once it's generated. • Sees themselves only under their anonymised site label.

[SCREENSHOT PLACEHOLDER]

Generate Summary button (idle)

GwasSummaryCard before clicking.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Loading state

Spinner / 'Generating...' state while the LLM runs.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Server log of outbound LLM call (optional)

Flask app terminal showing the POST to OpenAI being made — proves only aggregated digest is sent.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

T13 • Read & Share AI Report

End of the workflow

TIME	INITIATOR	COLLABORATION SERVER	COLLABORATOR
T13 End of the workflow	- Reads the final report: overview paragraph, ranked top SNPs, per-site contribution, recommendation callout. - GET /api/collaboration/<uuid>/gwas-summary returns the cached summary on later visits. - Can copy / export the report and discuss next steps with collaborators.	- Caches the report in collaborations.ai_summary; subsequent GETs are read-only. - Closes the workflow — no more compute is required unless the initiator regenerates.	- Opens CollaborationDetails and sees the same final report. - Identified only by site label (privacy preserved). - Can use the recommendation to decide on follow-up work.

[SCREENSHOT PLACEHOLDER]

Final AI Summary Card — top of report

Overview + Top SNPs section.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Final AI Summary Card — per-site analysis

The middle block of the summary card.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Recommendation banner (continue / caveats / reconsider)

The colored callout at the bottom of the card.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

[SCREENSHOT PLACEHOLDER]

Same report on Collaborator's screen

Confirms collaborators see only site labels, not real names.

Replace this box with your screenshot (right-click → Insert Picture, or paste).

How to capture local-machine screenshots

Some steps (T2, T7, T9) run on each party's own computer — the local QC controller at localhost:5001 does the actual data crunching. Here is how to capture evidence of that local activity so the placeholders in those sections can show real screenshots.

1. Terminal running qc_controller.py

On the machine that's playing the role (Initiator or Collaborator), start the controller with ``python qc_controller.py`` from `Collaborator_Server/`. Leave the Terminal window visible. Trigger the QC from the web UI and screenshot the Terminal as the log lines scroll by: 'Running MAF...', 'Surviving samples: N', etc. macOS: Cmd+Shift+4 + Space to capture just the window. Windows: Snip & Sketch (Win+Shift+S).

2. Browser DevTools → Network tab

Open the web app, hit F12 / Cmd+Option+I, switch to the Network tab, filter by `'/api/qc'` or `'/api/create_gwas_dataset'`. Trigger the action. Screenshot the network panel showing the request being POSTed to localhost:5001 — this proves the orchestrator is talking to the local controller, not pulling raw data into the cloud.

3. Raw data file in Finder / Explorer

Open the folder where the party's raw genotype CSV lives (e.g. `~/data/eye_color.csv`). Screenshot the Finder/Explorer window showing the file timestamp before and after the QC run is unchanged — emphasises that QC reads it but doesn't move it.

4. Flask app server log

On whichever machine the Collaboration Server is running, screenshot the Flask log lines showing the orchestrator firing requests to each party's QC controller URL. This is what proves the server is coordinating, not computing on raw data itself.

5. Side-by-side composite

For the most impressive figure, take two screenshots at the same moment — one of the Initiator's local Terminal and one of the Collaborator's local Terminal — and place them side by side in the doc. This visually demonstrates that both parties are crunching their own data in parallel.

How to drop multiple screenshots into one section

Each phase already has several dashed placeholder boxes. Click inside a box, delete the placeholder text, and use Insert → Pictures → This Device. If you need more boxes, copy any existing dashed box (click its border, Cmd/Ctrl+C, Cmd/Ctrl+V) and paste it below — Word preserves the formatting.

Notes on privacy and data handling

- Raw genotype data never leaves a party's local environment. Only QC survivor lists and per-site case/control counts are shared with the Collaboration Server.
- The chi-square aggregation uses contingency tables only — there is no row-level genotype exchange between Initiator and Collaborator.
- The AI summary is built from an already-aggregated digest. Collaborators are anonymised as Site A / Site B / ... in both the prompt and the rendered report; participant IDs are never sent to the LLM.
- The generated summary is cached server-side so that subsequent reads (GET /api/collaboration/<uuid>/gwas-summary) do not re-call the LLM.